

Assignment 2. Gauss-Jordan & LU

Marks 20

Posted on 19.08.2026 @ 2:30 pm and due on 19.08.2026 @ 5:40 pm

1. Prepare library routines for Gauss-Jordan elimination method using *row pivoting* and LU decomposition. For LU mention your choice of *Doolittle* or *Crout* clearly. [5+5]

2. Check Gauss-Jordan with the system of equations for three variables. [5]

$$\begin{array}{rcl} 2y + 5z & = & 1 \\ 3x - y + 2z & = & -2 \\ x - y + 3z & = & 3 \end{array}$$

3. Find LU decomposition of the matrix and verify. [5]

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{pmatrix}$$